Total No. of Questions : 10]

SEAT No. :

P-9345

[Total No. of Pages : 4

[6181]-580

B.E. (Mechanical/Mechanical Sandwich)
MECHANICAL SYSTEM DESIGN
(2015 Pattern) (Semester - II) (402048)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data if necessary.***
5) ***Use of logarithmic tables slide rule, Moiller charts, electronic pocket calculator and steam tables is allowed.***

Q1) a) What is structural formula Explain the procedure to construct the structural diagram for machine tool gear box. Draw structural diagram for following structural formulae 2(1) 2(2) 2(4) **[6]**

b) Compare between normal distribution and Standard normal distribution. **[4]**

OR

Q2) a) Why there is necessity of machine tool gear box for machine tool. Compare Structural Diagram & Speed Diagram. **[6]**

b) Two populations X and Y added together. Derive the expression to find mean of the resultant population. **[4]**

Q3) A triple ply belt conveyor is required to transport 4 tons of iron ore per hour at a conveyor speed of 3 m/s. If the mass density of iron ore is 2.5 tons/m^3 **[10]**

Determine :

a) Maximum suitable inclination for the conveyor

b) Diameter of the drive pulley

c) The ger box reduction ratio, if the motor speed is 1440 rpm.

P.T.O.

Use following data

Belt Inclination (α)	16°-20°	21°-25°	26°-30°	31°-35°
Flowability factor 'k'	2.5×10^{-4}	2.35×10^{-4}	2.20×10^{-4}	2.05×10^{-4}

Material factor for plies for belt, $K_1 = 2.0$

Belt tension and arc of contact factor, $K_2 = 80$

OR

Q4) a) Explain the need of idlers in belt conveyors, also draw different types of idlers used in belt conveyors. **[4]**

b) A horizontal belt conveyor is used for transporting the bulk material having mass density of 11772 N/m^3. The surcharge factor C for the flat belt is 0.2 while the belt width is 700 mm. If the belt speed is 1.75 m/s, determine the capacity of conveyor. **[6]**

Q5) a) Derive the expression to determine the principal stresses at the inner surface of a thick cylinder. **[8]**

b) A seamless cylinder of storage capacity 0.025 m^3 and made of alloy steel 20Mo55 (Sut = 500 N/mm^2) is to be used to store a fluid at 15 MPa gauge pressure. The length of the cylinder is twice its inner diameter. If the factor of safety is 2, determine the cylinder dimensions: **[10]**

OR

Q6) a) Explain with sketch the different types of formed heads used as end closures in cylindrical pressure vessels. Explain the design procedure of hemispherical head. **[8]**

b) A cylindrical pressure vessel is made up of stainless steel. Assuming the following data, determine the thickness of vessel shell

Internal diameter of vessel : 1500 mm

Design pressure: 0.44 MPa

Permissible stress of material : 130 MPa

Weld joint efficiency : 85 %

Assuming both ends as closed and external forces as well as corresponding stresses as negligible, determine the resultant stress in the shell. **[10]**

Q7) a) What are the types of piston rings? Why is more number of thin piston rings preferred over small number of thick rings? **[6]**

b) A 2 stroke petrol engine is to be designed for a brake power of 7kW at a speed of 800 rpm. The mechanical efficiency is 80%. If the indicated mean effective pressure is 0.5 MPa, determine: (i) the bore and length of cylinder liner; (ii) the thickness of cylinder liner; (iii) the thickness of cylinder head and (iv) Resultant stresses in the liner if the Poisson's ratio is 0.25. **[10]**

OR

Q8) The following data refers to a four stroke diesel engine : **[16]**

a) Cylinder bore = 250 mm,

b) Piston stroke = 300 mm,

c) Speed = 600 rpm,

d) Indicated mean effective pressure = 0.6 N/mm^2,

e) Maximum gas pressure = 4 N/mm^2,

f) Brake specific fuel consumption = 0.25 kg/kW–h

g) Higher calorific value of fuel = 44,000 kJ/kg

h) Mechanical efficiency = 80%.

Assume that 5% of total heat developed in the cylinder is transmitted by the piston. The piston is made of gray cast iron FG200 (Sut = 200 N/mm^2 and K = 46.6 W/m°C) and the factor of safety is 5. The temperature difference between the center and the edge of the piston head is 220°C

a) Determine the thickness of the piston head by strength consideration and thermal consideration.

b) State whether a cup is required in the top of piston head, If so calculate the radius of the cup.

Q9) a) Explain the Johnsons method of optimum design. **[6]**

b) A shaft is to be used to transmit a torque of 900 N-m. The required torsional stiffness of the shaft is 90 N-m/degree. While the factor of safety based on yield strength is 1.5. Using maximum shear stress theory, design the shaft with objective of minimizing the weight. **[10]**

Material	Mass density, ρ (Kg/m^3)	Modulus of rigidity, G (GPa)	Yield Strength Syt (MPa)	Material Cost, C (Rs/N)
M1	8500	80	130	16
M2	3000	26.5	50	32
M3	4800	40	90	480
M4	2100	16	20	32

OR

Q10) A thin spherical vessel is subjected to an internal pressure of 5 N/mm^2. The mass of the empty vessel should not exceed 150 kg. If the factor of safety based on yield strength is 2.0. Design the pressure vessel with the objective of maximizing the gas pressure out of the following materials. **[16]**

Material	Tensile Yield Strength Syt (MPa)	Mass density, ρ (Kg/m^3)
Alloy steel 15Cr90Mo55	450	7800
Al Alloy	150	2800
Titanium alloy	800	4500
Magnesium alloy	100	1800

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